Advanced ERM — Empirical IO Assessed Exercise 2026

Estimating Demand using Nested and Mixed Logit Models

Students may use `R` or any other language to answer this question. The code on the course website if used should be adapted to the specifics of the question.

Students should set seed to 130.

Answer the following questions by generating a PDF from an R Markdown file (or equivalent for another programming language) which includes your code, its output, and your answers to questions in a single document.

Where the computer algorithm has many iterations, use settings that do not print output at each iteration, in order to avoid a very long document.

If you experience numerical difficulties implementing what is requested, particularly when estimating the full BLP model, then it is acceptable to say so and to make specification adjustments to obtain something that works better.

Each time you are asked to comment it should be no more than a short paragraph.

The relative weighting given to each part of the question is indicated in square brackets.

1. *Beer demand*

   (a) Look at the simulated data provided on the course website, `beerData.csv`. The variables are

   - `product_id`, product identifier
   - `market_id`, market identifier
   - `num_prod_mkt`, number of products in market
   - `marketsize`, market size (i.e. the total number of potential purchase occasions in the market)
   - `quantity`, quantity in pints.
   - `price`, price in £.
   - `marginal_cost`, marginal cost in £.
   - `alc_j`, alcohol content

   Briefly describe the key variables in the data. [10%]

   (b) Consider a nested logit model where all the inside goods are in the same nest. Consumers care about two product characteristics (a constant term and beer strength) and the price. Calculate the market shares needed for the nested logit. Estimate the nested logit model using the inversion method. Discuss your identification assumptions. Using the residuals from your estimated equation create a vector, ξ, for later use, and interpret this as the unobserved utility of each product. Comment on your estimates and standard errors. (As a rough guide, the price parameter should be around -0.1 and the GEV nesting parameter should be around 0.8). [10%]

   (c) Calculate the own-price elasticities. Assume that firms set prices under a single-product Nash assumption. Back out the marginal costs using the inverse-elasticity rule (i.e. the single-product Nash assumption) and describe them. Comment on

your own-price elasticities and marginal costs. Is there much market power? Calculate the diversion ratio to the outside good and compare with the outside good market share. Comment on the flexibility of the model. [10%]

(d) The government is considering a £10 tax per unit of beer strength (`alc_j`). The beer industry is concerned about the loss of sales volume. Calculate the counterfactual prices after the tax is imposed using the nested logit model. Compare the average price increase with the average marginal cost increase and comment. Analyze the change in sales for different types of beers. What is the percentage change in the (unweighted) average price caused by the the tax? Evaluate the percentage effect on total quantity of beer sold. What does this suggest about the elasticity of aggregate demand for beer? (Tips. For the counterfactual, use the marginal costs you have backed out, rather than the ones that are given in the data. Also, as a check on your code, it is good to make sure that, when you solve for prices, they are the same as the observed prices in the case when there is no tax.) [20%]

2. *Auto demand*

(a) On the course website look at the market level car data `autoData.csv` and car model ownership data: `dummies_cars.csv`. Consider two alternative specifications of a BLP model of car demand with random coefficients. In the first specification the the utility for consumer i and product j in market t is

$$u_{ijt} = \sum_{k=1}^{5}(\beta_k + \sigma_k \nu_{ik})x_{jk} - (\alpha + \sigma_p \nu_{ip})p_{jt} + \xi_{jt} + \varepsilon_{ijt}$$

where β_k is the mean taste for the kth characteristic (x_{jk}), ν_{ik} is a standard normal taste shock which is iid across (i, k) and σ_k is a k-specific scaling term. ξ_{jt} is unobserved utility. α is the mean price parameter, σ_p is a scaling term and ν_{ip} is an iid standard normal effect for the price coefficient. ε_{ij} is a standard Type-1 Extreme Value random term which is iid across consumers and products. In the second specification the utility is

$$u_{ijt} = \sum_{k=1}^{5}(\beta_k + \sigma_k \nu_{ik})x_{jk} - \alpha p_{jt}/y_i + \xi_{jt} + \varepsilon_{ijt}$$

where all notation is as before and income y_i is drawn from the same log-normal distribution used in BLP(1995) and in the lecture: i.e. Lognormal(μ_t, σ_y^2) where $\sigma_y = 1.72$ and μ_t varies over time as given in the following table.

Use `BLPestimatoR` in R (or any other package/language of your choice) to estimate the two specifications using BLP instruments. Comment on whether you cluster when computing standard errors. Discuss how you chose the settings in the algorithm and whether your estimates varied for different starting values. Compare the two sets of estimates. [20%]

(b) For each of the two estimated models calculate own- and cross-price elasticities for 1971. Display these in a matrix for an interesting subset of the products.

Table 1: Values of μ_t from 1971 to 1990

Year	1971	1972	1973	1974	1975
μ_t	2.01156	2.06526	2.07843	2.05775	2.02915
Year	1976	1977	1978	1979	1980
μ_t	2.05346	2.06745	2.09805	2.10404	2.07208
Year	1981	1982	1983	1984	1985
μ_t	2.06019	2.06561	2.07672	2.10437	2.12608
Year	1986	1987	1988	1989	1990
μ_t	2.16426	2.18071	2.18856	2.21250	2.18377

Calculate marginal costs and markups for 1971. Compare marginal costs and markups for high price cars and low price cars under the two models and comment. Comment on how the two models differ. (Tip: update the 'mean utility' stored in the computer after estimation. To do this use update_BLP_data(). See the BLP vignette for details. As a check you have done this make sure your two models are predicting the same aggregate market shares as each other for any given product.) [10%]

(c) For 1971 calculate the effect of imposing a 20 percent VAT (sales tax) on prices. Use both models. A sales tax should be modeled such that, for product j, $p_j^c = 1.2p_j^f$ where p_j^c denotes the price the consumer pays and p_j^f denotes the price the firm receives. Consequently, the code for the first order condition for setting prices needs to be adjusted slightly. If you are doing this right it should not be a major change. Explain how you do this. Compare the original prices before the tax with the consumer and producer prices after the tax. Some commentators are concerned that the percentage effect on consumer prices will be higher for low-price than for high-price cars. Do the two models support this concern? Comment. [20%]

Advanced Empirical Research Methods - Production Functions

Assessment Exercise

Suppose we have a two-factor Cobb-Douglas production function for the log of value-added (v_{it})

$$v_{it} = \beta_0 + \beta_L l_{it} + \beta_K k_{it} + a_{it}$$

with $\beta_L + \beta_K < 1$. The labour input ($L_{it} = \exp(l_{it})$) is perfectly flexible and chosen in period t to maximise net revenue after the firm has observed total factor productivity ($A_{it} = \exp(a_{it})$) and both output and input prices in period t. The capital input ($K_{it} = \exp(k_{it})$) is predetermined and chosen in period $t-1$ to maximise the expected value of net revenue in period t conditional on information available to the firm in period $t-1$. Neither input is subject to any form of adjustment costs. The underlying production technology for gross output can be assumed to be Leontief in value-added and a perfectly flexible intermediate input (M_{it}).

Firms are price-takers in all markets. The price of one unit of value-added is common across firms, fixed over time, and normalised to one. Units of labour are hired at the wage W_{it} in period t. Units of capital are hired at the rental rate U in each period, which is common across firms and fixed over time. The firm's information set in period $t-1$ includes the fixed rental rate for capital, the fixed price of one unit of value-added, total factor productivity in period $t-1$ ($A_{i,t-1}$), and the wage in period $t-1$ ($W_{i,t-1}$), but does not include total factor productivity in period t (A_{it}) or the wage in period t (W_{it}).

The log of the wage for labour ($w_{it} = \ln W_{it}$) varies across firms and over time in a way which is independent across firms but may be persistent over time; more specifically, we have $w_{it} = w + c_{it} + \phi c_{i,t-1}$ with $c_{it} \sim \text{i.i.d.N}(0, \sigma_c^2)$, so that for each firm the log wage follows a non-trivial MA(1) process if $\phi \neq 0$.

The log of total factor productivity (a_{it}) follows a non-trivial AR(1) process

$$a_{it} = \rho a_{i,t-1} + \xi_{it}$$

with $0 < \rho < 1$ and $\xi_{it} \sim \text{i.i.d.N}(0, \sigma_\xi^2)$.

The available data on the log of value-added ($\widetilde{v}_{it}$) is measured with an additive error, so that we observe

$$\widetilde{v}_{it} = v_{it} + \varepsilon_{it}$$

with $\varepsilon_{it} \sim \text{i.i.d.N}(0, \sigma_\varepsilon^2)$.

Data is observed for the output and input quantities ($\widetilde{V}_{it} = \exp(\widetilde{v}_{it}), L_{it}, K_{it}$) only.

1. Suppose we know that $\phi = 0$.

   (a) Outline a two stage procedure that could be used to estimate the parameter β_L consistently from the first stage specification, and to estimate the parameter β_K consistently from the second stage specification.

   (b) Provide code in R which generates simulated production data for this case for many time periods for one firm, and implements the estimation method that you have outlined using the simulated data.

   (c) Is the second stage of this procedure needed to estimate the parameter β_K consistently in this setting? If so, explain why. If not, explain which features of the data generating process considered here allow both of the output elasticity parameters to be estimated consistently from the first stage specification.

2. A researcher suggests that the two stages of this estimation procedure could be combined by considering a specification of the form

$$\widetilde{v}_{it} = \beta_0 + \beta_L l_{it} + \beta_K k_{it} + \rho a_{i,t-1} + \xi_{it} + \varepsilon_{it}$$

   using a control function for $a_{i,t-1}$ of the form $a_{i,t-1} = h(k_{it}, k_{i,t-1})$ and a two stage least squares (2SLS) estimator with $l_{i,t-1}$ used as an outside instrument for l_{it}.

   (a) Explain why we would require an outside instrument for l_{it} in this specification.

   (b) Could we estimate β_L consistently using this 2SLS estimator if $\phi = 0$? If so, explain why. If not, explain the nature of the problem. Provide code in R which uses simulated production data to illustrate your answer.

   (c) Could we estimate β_L consistently using this 2SLS estimator if $\phi \neq 0$? If so, explain why. If not, explain the nature of the problem. Provide code in R which uses simulated production data to illustrate your answer.

   (d) Could we estimate β_K consistently using this specification, even if $\phi = 0$ and the value of β_L is known? If so, explain why. If not, explain the nature of the problem. Provide code in R which uses simulated production data to illustrate your answer.

Advanced Empirical Research Methods

Marginal Treatment Effect

Exam rules

Use `R` and produce your answers by generating a PDF from an `R Markdown` file which includes your code, its output, and your answers to questions in a single document.

Please, be concise in your answers: each time you are asked to comment, it should be no more than two short paragraphs, unless specified otherwise. Wordiness that betrays a lack of understanding will be penalized.

Do this exam on your own. Do not copy other student's code. If this happens, both students will fail the exam. So please do not ask a fellow student for this kind of favor.

You can use code from your homework, build on ChatGPT, etc., and use built-in functions, but your code must demonstrate that you understand what is happening. Please refer to the guidance on AI on the Canvas page.

Finally, if you find that the exam is missing assumptions to solve a problem, make assumptions that will allow you to proceed, and state clearly what these assumptions are.

Exam Question

This exam is based on the paper Heckman, Urzua and Vytlacil (2006), which describes several options for estimating marginal treatment effects.

(Q1) Set the seed to 34837 and simulate a sample of size $N = 5{,}000$ from the example in Section 10 of the paper. Estimate the propensity score and check that it has full support. In your answer, explain what full support means in this setting and why it matters for marginal treatment effect estimation.

(Q2) Explain how we arrive at the equation at the bottom of page 4, starting from the latent-index model and the joint normality assumption. Use the parametric methodology in Section 5.1 to estimate the MTE. Provide a graphical illustration in the spirit of Figure 4. Your answer should clearly explain the steps used to obtain the estimated MTE.

(Q3) Use the methodology in Section 5.2 to estimate the MTE. Assess the sensitivity of your estimates to the choice of the polynomial degree. Briefly comment on the differences.

(Q4) Use the methodology in Section 7 to estimate the MTE. Carefully explain your choices of procedure for Step 2 and Step 3. Provide the resulting graph and compare it with the parametric and polynomial estimates.

(Q5) Provide an alternative simulated data-generating process in which the normality assumption is violated. Use your three estimators from Questions 2, 3, and 4 to represent the MTE on this new dataset. Compare the three curves and comment on what changes when normality fails. Your answer should make clear which features of the MTE are robust and which features are sensitive to the parametric assumption.